

Impulsive Noise Mitigation in Underwater Acoustic OFDM Systems

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Abstract—Mitigation of impulsive noise has been extensively studied in wireline, wireless radio, and powerline communication systems. However, its study in underwater acoustic (UWA) systems is quite limited. This paper considers impulsive noise mitigation for underwater orthogonal frequency-division multiplexing (OFDM) systems, where the system performance is severely impacted by the channel Doppler effect. We propose a practical approach based on a least squares formulation: First, the positions of impulsive noise are determined in the time domain based on the signal amplitude, and second, impulsive noise samples are jointly estimated with the Doppler shift based on the measurements of the OFDM null subcarriers. Based on the available channel estimate and tentative data symbol decisions, an iterative receiver is further developed. Data sets have been acquired in a recent sea experiment near Kaohsiung city, Taiwan, in May 2013. Performance results based on extensive simulations and collected data sets demonstrate that the proposed receivers effectively mitigate impulsive noise for UWA OFDM systems.

Index Terms—Impulsive noise, iterative receiver, null subcarrier, underwater acoustic (UWA), orthogonal frequency-division multiplexing (OFDM).

I. INTRODUCTION

IMPULSIVE noise mitigation has been extensively studied in the literature of digital subscriber line, wireless communications, and powerline communications (see, e.g., [1]–[9]). One traditional method to reduce the effect of impulsive noise is to precede the demodulator with memoryless nonlinearity. Popular types of memoryless nonlinearity include clipping, blanking, and combined clipping–blanking (see a comparative study of these three nonlinearities in [6]). After the clipping or blanking processing, iterative receiver processing

that relies on impulsive noise reconstruction and cancellation after tentative data decoding can further improve system performance [4], [10], [11].

Another representative method is to apply compressive sensing (CS) techniques to estimate impulsive noise directly by using the measurements on null subcarriers of orthogonal frequency-division multiplexing (OFDM) modulation [12], [13]. The proposed method was improved in [14] by exploiting the structure of the sensing matrix. Recently, a factor-graph-based receiver is proposed in [15] to perform joint channel, impulsive noise, symbol, and bit detection via message-passing techniques.

This paper considers an underwater acoustic (UWA) OFDM system, which has recently become a popular choice for underwater networks. Note that the UWA channel is rich in interference, such as interference from sonar operations, marine mammals, shrimp snapping noise, and malicious jamming [16]. The impact of shrimp snapping noise on the performance of underwater OFDM has been reported in [17] and [18], and a receiver design to mitigate partial-band partial-block-duration interference, e.g., due to concurrent sonar users, has been recently proposed in [19]. The presence of impulsive noise is often observed in field experiments [16], [17], [20]–[26]. Typical causes of impulsive noise are as follows. The biological sources of noise, such as snapping shrimp, which live in certain geographical areas, often produce high-level impulsive noise [22], [23]. In the near shores and in the presence of shipping activity, manmade noise significantly increases the noise level [27]. Industrial noise associated with oil and gas exploration and production, such as explosions and pile drivers, is another source for impulsive noise [27].

Due to the slow propagation speed of sound, platform motion and instability of water medium lead to waveform compression/dilation, creating frequency-dependent Doppler shifts and fast channel variations. One primary task of the receiver design for UWA OFDM is to deal with the severe Doppler effects and channel variations, e.g., [28]–[33]. So far, the two-step Doppler mitigation approach, which is a resampling step to remove the major Doppler scaling effect followed by high-resolution compensation on the residual Doppler shift, has been shown effective in typical situations [28]. Notably, the null subcarrier method in [34]–[36] has been adopted to estimate the residual Doppler shift [28], which is implemented in OFDM acoustic modem prototypes [37].

Existing studies of OFDM in the presence of impulsive noise have assumed either an additive white Gaussian noise (AWGN) channel or static multipath channels [1]–[13]. Those methods may not be directly applicable to UWA systems as the channel

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Doppler effects have to be dealt with explicitly. The main contributions of this work are as follows.

- We develop an effective approach for joint Doppler shift and impulsive noise estimation based on a least squares (LS) formulation. The positions of the impulsive noise are determined based on the signal amplitude in the time domain, whereas the estimation of Doppler shift and impulsive noise samples relies on the measurements on OFDM null subcarriers. With available channel estimates and tentative symbol decisions, we further develop an iterative receiver.
- We have obtained useful experimental data sets where the received OFDM blocks contain impulsive noise. Performance results based on the data sets and simulations with real impulsive noise characteristics validate that the proposed receivers are very effective. In particular, the iterative receiver can approach the benchmark performance whereby impulsive noise is absent.

A consideration of Doppler shift and impulsive noise together is only available in this work and our preliminary results in [38] and [39]. The differences of this work from that in [38] and [39] are as follows. 1) This work presents a new approach based on the LS formulation, whereas that in [38] and [39] is based on CS. 2) Real experimental data are used in this work, which were not available in [38] and [39]. 3) Simulations in this work use the noise characteristics derived from real data, which is not the case in [38] and [39]. 4) This paper adopts a zero-padded (ZP) OFDM presentation to be consistent with the experimental settings, whereas cyclic-prefix OFDM is used in [38] and [39]. Finally, performance results based on recorded data sets and simulations with real noise characteristics show that the LS-based approach outperforms the CS-based approach in [38] and [39] considerably.

The rest of this paper is organized as follows. Section II describes the system model. Section III presents the proposed noniterative and iterative methods. Simulation results are presented in Section IV, whereas Section V contains performance results using experimental data sets. Conclusions are drawn in Section VI.

Notation: Bold uppercase and lowercase letters are used to denote matrices and column vectors, respectively. $(\cdot)^T$ and $(\cdot)^H$ denote transpose and Hermitian transpose, respectively. $\mathbf{I}_N$ denotes an identity matrix of size $N \times N$.

II. SYSTEM MODEL

UWA signals have low frequencies, e.g., below 100 kHz, and passband signals are directly used for signal generation and recording. As such, continuous passband signals and channels are often used for system modeling. We follow the formulation in [28] for a ZP OFDM system.

Let T denote the OFDM symbol duration and T_g the length of the guard interval. The total OFDM block duration is $T_{\text{bl}} = T + T_g$. The subcarrier spacing is $\Delta f = 1/T$. A total of K subcarriers are located at the following frequencies:

$$f_k = f_c + k\Delta f, \quad k = -\frac{K}{2}, \dots, \frac{K}{2} - 1 \quad (1)$$

where f_c is the center frequency. The signal bandwidth is thus $B = K/T$. Define $\mathcal{S}_A$ and $\mathcal{S}_N$ as the nonoverlapping sets of active subcarriers and null subcarriers, respectively, which satisfy $\mathcal{S}_A \cup \mathcal{S}_N = \{-K/2, \dots, K/2 - 1\}$. Let $d[k]$ denote the information symbol on the k th subcarrier. The transmitted signal of one OFDM block in passband can be expressed as

$$\tilde{x}(t) = 2\text{Re} \left\{ \left[\sum_{k \in \mathcal{S}_A} d[k] e^{j2\pi k \Delta f t} g(t) \right] e^{j2\pi f_c t} \right\}, \quad t \in [0, T_{\text{bl}}] \quad (2)$$

where $g(t)$ is the zero-padding operation as

$$g(t) = \begin{cases} 1, & t \in [0, T] \\ 0, & t \in [T, T_{\text{bl}}]. \end{cases} \quad (3)$$

A time-varying UWA channel can be represented by

$$h(\tau, t) = \sum_p A_p(t) \delta(t - \tau_p(t)) \quad (4)$$

where $A_p(t)$ and $\tau_p(t)$ are the amplitude and the delay of the p th path, respectively. As in [28], we adopt the following assumptions.

A1) All paths have a similar Doppler scaling factor a such that

$$\tau_p(t) \approx \tau_p - at. \quad (5)$$

In general, different paths could have different Doppler scaling factors. The receiver in this paper is based on the assumption that all the paths have the same Doppler scaling factor for simplicity. When this is not the case, part of useful signals is treated as additive noise, which could increase the overall noise variance considerably, or a more complex receiver is needed, e.g., [31] and [32], which can explicitly deal with path-specific Doppler scales.

A2) The path delays τ_p , the gains A_p , and the Doppler scaling factor a are constant over the block duration T_{bl} .

The OFDM block duration used in the experiment is $T_{\text{bl}} = 250$ ms. Assumption A2) is reasonable within this duration, as the channel coherence time is usually on the order of seconds.

Based on assumptions A1 and A2, the received passband signal in the presence of impulsive noise is then

$$\tilde{y}(t) = 2\text{Re} \left\{ \sum_p A_p \left[\sum_{k \in \mathcal{S}_A} d[k] e^{j2\pi k \Delta f (t+at-\tau_p)} g(t+at-\tau_p) \right] \times e^{j2\pi f_c (t+at-\tau_p)} \right\} + \tilde{i}(t) + \tilde{w}(t) \quad (6)$$

where $\tilde{i}(t)$ is impulsive noise, and $\tilde{w}(t)$ is ambient noise. The following operation is performed on the received signal [28]. We resample the received signal $\tilde{y}(t)$ by using a resampling factor $\hat{a}$, which leads to the resampled signal as $\tilde{y}(t/(1+\hat{a}))$.¹

¹Estimation of the resampling factor is done typically by the use of preamble and postamble signals [40] or a preamble alone [41]. Since there is no information transmission, estimation of a single parameter is much more robust than data transmission, even when impulsive noise occurs.

After downshifting and lowpass filtering (LPF), we obtain the baseband signal, i.e.,

$$y(t) = \text{LPF} \left[\tilde{y} \left(\frac{t}{1 + \hat{a}} \right) e^{-j2\pi f_c t} \right] \\ \approx e^{j2\pi \epsilon t} \sum_{k \in \mathcal{S}_A} \left\{ d[k] e^{j2\pi k \Delta f t} \left[\sum_p A_p e^{-j2\pi f_k \tau_p} g(t - \tau_p) \right] \right\} \\ + i(t) + w(t) \quad (7)$$

where $\epsilon = ((a - \hat{a})/(1 + \hat{a}))f_c$, and $i(t)$ and $w(t)$ are impulsive and ambient noise samples in the baseband.

Sampling $y(t)$ at the baseband rate $1/B$, we obtain the discrete samples $y[n] = y(t)|_{t=n/B}$. For each block, a total of $(K + N_g)$ samples are obtained, where $N_g = \lfloor KT_g/T \rfloor$.

To obtain an equivalent representation, define a rectangular window as $g[n] = 1$ for $n = 0, \dots, K - 1$ and zero elsewhere. Define the channel frequency response as

$$C(f) = \sum_p A_p e^{-j2\pi f \tau_p}. \quad (8)$$

The discrete-time baseband channel has a finite number of taps, with the l th tap defined by [42]

$$h[l] = \sum_{k=-K/2}^{K/2-1} C(f_k) e^{\frac{j2\pi l k}{K}}. \quad (9)$$

This way, the frequency-domain response of the baseband channel $\{h[l]\}$ matches that of the continuous-time multipath channel at the subcarrier frequencies [42]. With the discrete-time channel, the received data samples can be expressed as

$$y[n] = e^{\frac{j2\pi \epsilon n}{B}} \sum_l h[l] \sum_{k \in \mathcal{S}_A} d_k e^{\frac{j2\pi k(n-l)}{K}} g[n-l] + i[n] + w[n] \quad (10)$$

where $i[n]$ and $w[n]$ are the impulsive and ambient noise samples.

Assume that the channel has $L + 1$ taps in the discrete time, and $L \leq N_g$. Hence, a total of $K + L$ samples contain all the useful information about the current block. Define the following vectors:

$$\mathbf{h} := [h[0], \dots, h[L]]^T \\ \mathbf{y} := [y[0], \dots, y[K + L - 1]]^T \\ \mathbf{i} := [i[0], \dots, i[K + L - 1]]^T \\ \mathbf{w} := [w[0], \dots, w[K + L - 1]]^T$$

and a matrix

$$\mathbf{\Lambda}(\epsilon) := \text{diag} \left(1, e^{\frac{j2\pi \epsilon}{B}}, \dots, e^{\frac{j2\pi \epsilon (K+L-1)}{B}} \right). \quad (11)$$

Define $\mathbf{X}$ as a $(K + L) \times L$ lower triangular Toeplitz matrix with first column $[x[0], x[1], \dots, x[K - 1], 0, \dots, 0]^T$, and the element is expressed as

$$x[i] = \sum_{k \in \mathcal{S}_A} d_k e^{\frac{j2\pi k i}{K}}, \quad i = 0, 1, \dots, K - 1. \quad (12)$$

The matrix-vector representation of the channel input-output relationship is

$$\mathbf{y} = \mathbf{\Lambda}(\epsilon) \mathbf{X} \mathbf{h} + \mathbf{i} + \mathbf{w}. \quad (13)$$

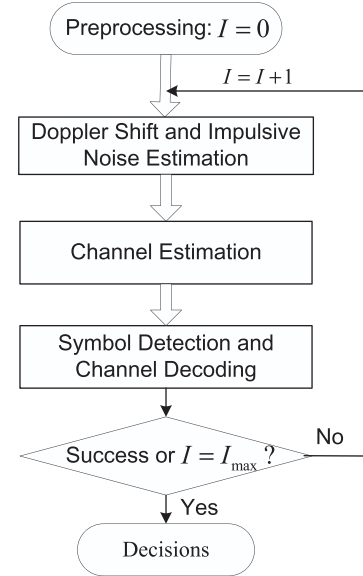


Fig. 1. Proposed receiver structure.

III. PROPOSED APPROACH

With the system model in (13), the Doppler shift and the impulsive noise need to be dealt with before carrying out channel estimation and data decoding. Based on the existing literature, the following methods are readily available.

- A receiver can use the clipping and blanking operation in the time domain to mitigate the impulsive noise [6] and then carry out Doppler shift compensation subsequently. This sequential approach is based on the large amplitude property to mitigate the impulsive noise.
- The Doppler shift and impulsive noise can be carried out either sequentially or jointly based on the null subcarrier measurements by using the CS method as proposed in [38] and [39]. This approach is based on an assumed sparsity of the impulsive noise.

Recent receivers based on sparse Bayesian learning [9] or factor graph [15] are not directly applicable to the system model in (13), as the Doppler shift is not considered in the receiver design.

Next, we present an LS-based approach, where the positions of impulsive noise are marked in the time domain based on the large amplitude property, and the impulsive noise samples are estimated together with the Doppler shift based on the frequency-domain measurements on the null subcarriers. An iterative receiver is then further developed based on tentative channel and symbol estimates. The diagram of the iterative receiver is shown in Fig. 1, which reduces to a noniterative receiver if the maximum number of iterations is set to zero.

A. Noniterative Processing

For the received vector $\mathbf{y}$, the receiver finds the median value of its $K + L$ samples as m_y and marks the positions of possible impulsive noise samples as

$$I[n] = \begin{cases} 1, & |y[n]| > m_y \Gamma \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

where Γ is a relative threshold [5]. Assume that there are N_i positions $n_1, \dots, n_{N_i}$ marked. Define a $(K + L) \times N_i$ position selection matrix $\mathbf{P}$ with unit entry at the (n_j, j) th position ($j = 1, \dots, N_i$) and zeros elsewhere. This way, the impulsive noise $\mathbf{i}$ is replaced by a parameterized version as

$$\tilde{\mathbf{i}} = \mathbf{P}\mathbf{e} \quad (15)$$

where the impulsive noise samples at the estimated positions are collected in an $N_i \times 1$ vector $\mathbf{e}$.

Joint estimation of ϵ and $\mathbf{e}$ is based on the frequency-domain samples. Define a matrix $\mathbf{O} = [\mathbf{I}_K \ \mathbf{I}_L]$ that performs an overlap-add operation on a vector, where $\mathbf{I}_L$ is the first L columns of $\mathbf{I}_K$. Define a $K \times K$ Fourier transform matrix $\mathbf{F}$ with the $(p + 1, q + 1)$ entry $e^{-j2\pi pq/K}$. Denote the indexes of the null subcarriers as $\{p_1, \dots, p_{K_1}\}$, where K_1 is the number of null subcarriers used. Define a $K_1 \times K$ subcarrier selection matrix $\mathbf{\Theta}$ with unit entry $\mathbf{\Theta}[k, p_k] = 1$ for $k = 1, \dots, K_1$, and zeros elsewhere. Incorporating the Doppler compensation and impulsive noise mitigation steps, the frequency-domain samples on the null subcarriers can be expressed as

$$\begin{aligned} \mathbf{z}_{\text{null}} &= \mathbf{\Theta}\mathbf{F}\mathbf{O}\mathbf{\Lambda}^H(\epsilon)(\mathbf{y} - \mathbf{P}\mathbf{e}) \\ &= \underbrace{\mathbf{\Theta}\mathbf{F}\mathbf{O}\mathbf{\Lambda}^H(\epsilon)}_{:=\mathbf{A}(\epsilon)}\mathbf{y} - \underbrace{\mathbf{\Theta}\mathbf{F}\mathbf{O}\mathbf{\Lambda}^H(\epsilon)\mathbf{P}}_{:=\mathbf{B}(\epsilon)}\mathbf{e}. \end{aligned} \quad (16)$$

Minimization of the energy on the null subcarriers leads to an optimization problem as

$$(\hat{\epsilon}, \hat{\mathbf{e}}) = \arg \min_{\epsilon, \mathbf{e}} \|\mathbf{A}(\epsilon)\mathbf{y} - \mathbf{B}(\epsilon)\mathbf{e}\|^2. \quad (17)$$

For each tentative ϵ , the corresponding $\mathbf{e}(\epsilon)$ that minimizes the objective function is

$$\mathbf{e}(\epsilon) = [\mathbf{B}^H(\epsilon)\mathbf{B}(\epsilon)]^{-1} \mathbf{B}^H(\epsilon)\mathbf{A}(\epsilon)\mathbf{y}. \quad (18)$$

Through a 1-D search on ϵ with $\hat{\mathbf{e}}(\epsilon)$ substituted into (17) to evaluate the objective function, the final estimate of $\hat{\epsilon}^{\text{opt}}$ is obtained. The impulsive noise estimate is then $\hat{\mathbf{e}}^{\text{opt}} = \mathbf{e}(\hat{\epsilon}^{\text{opt}})$ from (18).

The method in (17) is termed as the joint Doppler shift and impulsive noise estimation based on a nonlinear LS formulation. A special case is the sequential method where the Doppler shift is estimated without considering the presence of impulsive noise as in [28], and then, the impulsive noise vector is directly obtained as the LS solution in (18).

With the Doppler shift estimate $\hat{\epsilon}$ and the impulsive noise estimate $\hat{\mathbf{e}}$, the output of the useful frequency measurement on all subcarriers can be expressed as

$$\hat{\mathbf{z}} = \mathbf{F}\mathbf{O}\mathbf{\Lambda}^H(\hat{\epsilon})(\mathbf{y} - \mathbf{P}\hat{\mathbf{e}}). \quad (19)$$

Existing OFDM receiver processing can be performed on $\hat{\mathbf{z}}$, including channel estimation and data decoding. Specifically, the components of $\hat{\mathbf{z}}$ on the pilot subcarriers are used for channel estimation, and the measurements on the data subcarriers are used by a Viterbi decoder for decoding, following the approach in [28].

B. Iterative Processing

An iterative receiver can further improve the system performance after the initial processing by a noniterative receiver, as shown in Fig. 1. At the beginning of the k th iteration, the Doppler shift estimate $\hat{\epsilon}^{(k-1)}$, the channel estimate $\hat{\mathbf{h}}^{(k-1)}$, the symbol vector $\hat{\mathbf{d}}^{(k-1)}$, and, hence, the $\hat{\mathbf{X}}^{(k-1)}$ data matrix, from the $(k - 1)$ th iteration, are available and can be used to improve the Doppler shift and impulsive noise estimation.

First, construct a noise vector as

$$\mathbf{y} - \mathbf{\Lambda}(\hat{\epsilon}^{(k-1)})\hat{\mathbf{X}}^{(k-1)}\hat{\mathbf{h}}^{(k-1)} \quad (20)$$

based on which the positions of the impulsive noise will be updated using the thresholding method in (14). The updated position selection matrix is denoted as $\mathbf{P}^{(k)}$.

The joint Doppler shift and impulsive noise estimation is then formulated as

$$(\hat{\epsilon}^{(k)}, \hat{\mathbf{e}}^{(k)}) = \arg \min_{\epsilon, \mathbf{e}} \|\mathbf{y} - \mathbf{\Lambda}(\epsilon)\hat{\mathbf{X}}^{(k-1)}\hat{\mathbf{h}}^{(k-1)} - \mathbf{P}^{(k)}\mathbf{e}\|^2. \quad (21)$$

Solving (21) follows the same steps as solving (17). The frequency-domain samples at all subcarriers in the k th iteration are then updated as

$$\hat{\mathbf{z}}^{(k)} = \mathbf{F}\mathbf{O}\mathbf{\Lambda}^H(\hat{\epsilon}^{(k)})(\mathbf{y} - \mathbf{P}^{(k)}\hat{\mathbf{e}}^{(k)}) \quad (22)$$

for channel estimation and data decoding. The iterative operation stops once the information symbols have been successfully decoded (e.g., through the cyclic redundancy code bits embedded in the information symbols) or a fixed number of iterations has been reached.

C. Discussion

We first discuss the implementation complexity of the proposed receivers.

- The 1-D search on ϵ depends on the number of tentative values to be examined, denoted by N_ϵ . In a typical implementation, a fine-grid linear search would be replaced by a coarse-grid linear search followed by a bisectional search, which halves the search step size for each additional step. For the digital-signal-processing-based implementation of Doppler shift compensation in the absence of impulsive noise, N_ϵ is between 10 and 20 [37].
- For each tentative ϵ , the matrix $\mathbf{B}(\epsilon)$ is of size $K_1 \times N_i$. Assuming that the main complexity of the LS solution in (18) is due to the inversion of the matrix of size $N_i \times N_i$, which has a complexity on the order of $\mathcal{O}(N_i^3)$. For a small N_i , direct matrix inversion is affordable. For a large N_i , an iterative optimization algorithm, such as the least squares QR (LSQR) algorithm [43], can be used to find the LS solution.

In summary, the complexity of the LS-based joint method is on the order of $\mathcal{O}(N_\epsilon N_i^3)$, which reduces to $\mathcal{O}(N_i^3)$ for the LS-based separate method. The complexity of the iterative method scales linearly with the number of iterations. Note

TABLE I
OFDM PARAMETERS IN THE TAIWAN EXPERIMENT

Center frequency	f_c	17 kHz
Bandwidth	B	6 kHz
# of subcarriers	K	1024
Symbol duration	T	170.7 ms
Subcarrier spacing	Δf	5.88 Hz
Guard interval	T_g	79.3 ms
Convolution code rate	r	1/2

that the LS-based methods have lower complexity than the CS-based methods (e.g., based on the orthogonal matching pursuit implementation as in [38] and [39]), which need to identify both the positions and the noise sample amplitudes based on the same system model. A further comment is that UWA communication systems operate on the level of several kilobits per second with a system bandwidth of several kilohertz. Iterative receivers can be well accommodated for UWA systems, and there are numerous iterative receivers developed, e.g., [44]–[47].

Second, we discuss the potential impact of inaccurate position estimation of impulsive noise samples. As shown in (14), the positions are marked based on the large-amplitude property. For an OFDM signal, there exist large-amplitude signal samples due to the large peak-to-average power ratio. Although being marked, these large-amplitude signal samples might not be compromised as long as the noise estimates at those positions are small. On the other hand, the impulsive noise samples lower than the threshold will be ignored and treated as ambient noise. The impact of imperfect position estimation is evaluated by simulation in Section IV-C, where a performance loss of 1–2 dB can be seen. A further comment is that the positions of impulsive noise are estimated only based on the large-amplitude property, and there is no other constraint or other implicit models on the impulsive noise samples. Hence, the proposed receivers can work with impulsive noise samples, which are bundled together, as long as they have large amplitudes.

IV. SIMULATION RESULTS

The ZP-OFDM signal parameters are listed in Table I, which are used in the field experiment as described in Section V. The total 1024 subcarriers comprise 672 data subcarriers, 256 pilot subcarriers, and 96 null subcarriers. Out of the 96 null subcarriers, 24 null subcarriers are placed on each edge of the signal band, and 48 null subcarriers are mixed with the data and pilot subcarriers. The data symbols are drawn from a quaternary phase-shift keying constellation, and a 64-state rate-1/2 convolutional code is used for channel coding. Accounting for the overhead of pilot and null subcarriers, the guard zeros, and channel coding, the overall data rate is

$$R = \frac{1}{2} \cdot \frac{672 \cdot \log_2 4}{T + T_g} = 2.69 \text{ kb/s.} \quad (23)$$

The bit error rate (BER) after Viterbi decoding will be used as the performance metric.

The multipath channel used in the simulation consists of ten discrete paths,² where the interarrival time follows an exponential distribution with a mean of 1 ms. The path amplitudes are Rayleigh distributed with average power exponentially decreasing with the delay, where the difference between the beginning and the end of the guard time is 35 dB. The Doppler shift is randomly generated $\epsilon \in [-\Delta f/2, \Delta f/2]$, where Δf is subcarrier spacing.

Based on the system model in (13), the signal-to-noise ratio (SNR), the signal-to-impulsive-noise ratio (SIR), and the impulse-to-noise ratio (INR) at the receiver are defined as

$$\text{SNR} = \frac{\mathbb{E}[\|\mathbf{X}\mathbf{h}\|^2]}{\mathbb{E}[\|\mathbf{w}\|^2]}, \quad \text{SIR} = \frac{\mathbb{E}[\|\mathbf{X}\mathbf{h}\|^2]}{\mathbb{E}[\|\mathbf{i}\|^2]}, \quad \text{INR} = \frac{\mathbb{E}[\|\mathbf{i}\|^2]}{\mathbb{E}[\|\mathbf{w}\|^2]}.$$

A. Noise Generation

The composite noise $u[n] = w[n] + i[n]$ in (13) is generated based on the two-component Gaussian mixture (GM) model, which has been widely used for studying the performance of impulsive noise mitigation [6], [11], [13], [48]. The probability density function of this model is

$$f(u[n]) = (1-p)\mathcal{N}(u[n]; 0, \sigma_w^2) + p\mathcal{N}(u[n]; 0, \sigma_w^2 + \sigma_g^2) \quad (24)$$

where $\mathcal{N}(\cdot)$ is the complex Gaussian distribution function, σ_w^2 is the additive white Gaussian noise variance, σ_g^2 is the variance of the Gaussian component of impulsive noise, and p is the probability of occurrence of impulsive noise.

The model parameters p , σ_g^2 , and σ_w^2 can be derived from the noise signals recorded from a real field. Based on the impulsive noise data sets from the experiment described in Section V, Fig. 2(a) shows the baseband noise signal, and Fig. 2(b) shows that a two-component Rayleigh distribution model fits the noise empirical distribution using the estimated parameters, which are $p = 0.02$, $\sigma_w = 0.0011$, and $\sigma_g = 0.0216$. Fig. 3 plots a noise realization of the noise sequences $\{u[n]\}$ generated by the two-component GM model within one OFDM block duration, where the impulsive noise power is 26 dB above the background noise power.

In our following simulations, we keep $p = 0.02$ and vary σ_w and σ_g to have different SIR, SNR, and INR settings, except in Section IV-E where the value of p is also changed. When $p = 0.02$, about 20 impulsive noise samples are generated within one OFDM block.

B. Threshold Selection

In an AWGN channel, the work in [5] has tried to find the optimal clipping threshold based on theoretical analysis. Here, we determine the threshold numerically based on our problem setup. Fig. 4 shows the BER under two different settings: a) SIR = −10 dB, $p = 0.02$, SNR = 15 dB and b) INR = 26 dB, $p = 0.02$, SNR = 15 dB, respectively. In the first receiver, the impact of the thresholds used in the clipping–blanking method

²As the number of paths increases, one expects that the channel estimation accuracy decreases, and hence, the overall system performance deteriorates. The impact of channel length on the system performance is not within the scope of this paper.

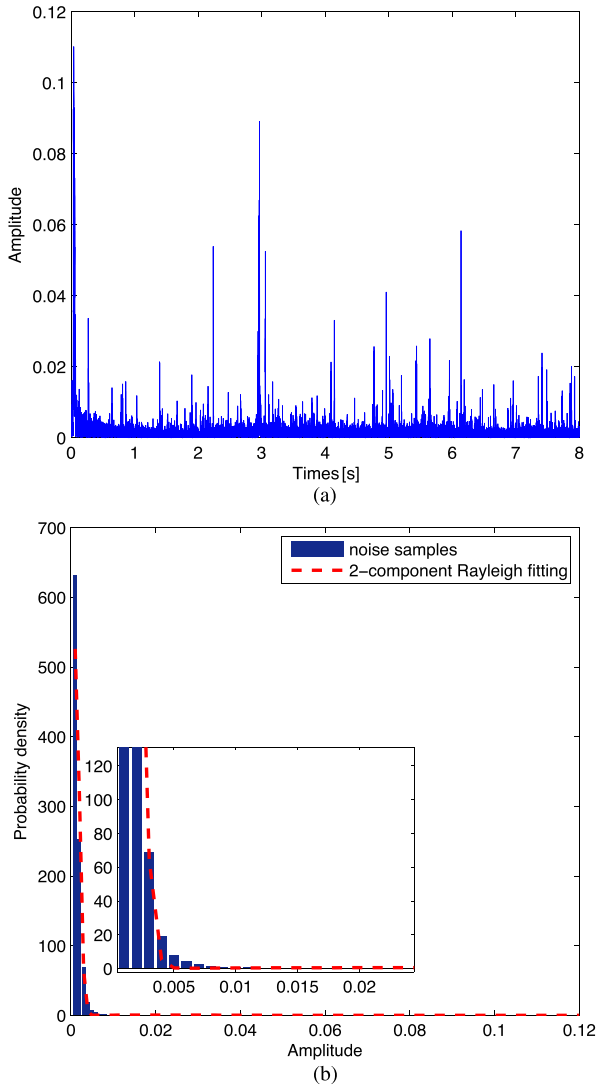


Fig. 2. Received baseband ambient noise signal and its probability distribution. (a) Baseband ambient noise signal waveform. (b) Two-component Rayleigh distribution fit for the amplitude of received baseband noise signal.

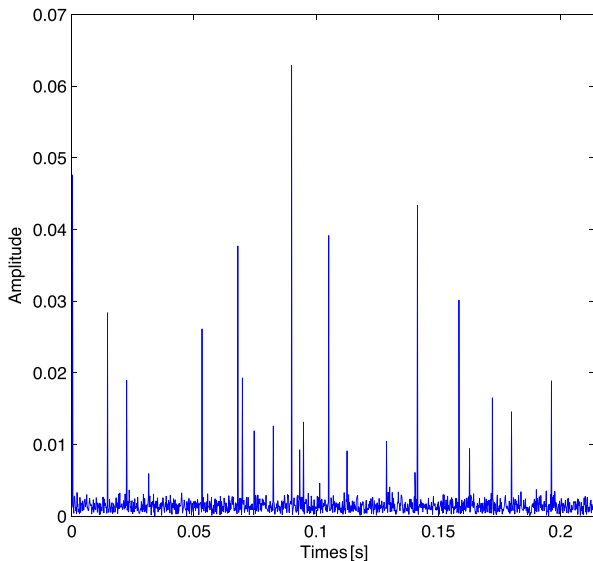


Fig. 3. Realization of total noise series $\{u[n]\}$ with the two-component GM model, $p = 0.02$, $\sigma_w = 0.0011$, and $\sigma_g = 0.0216$.

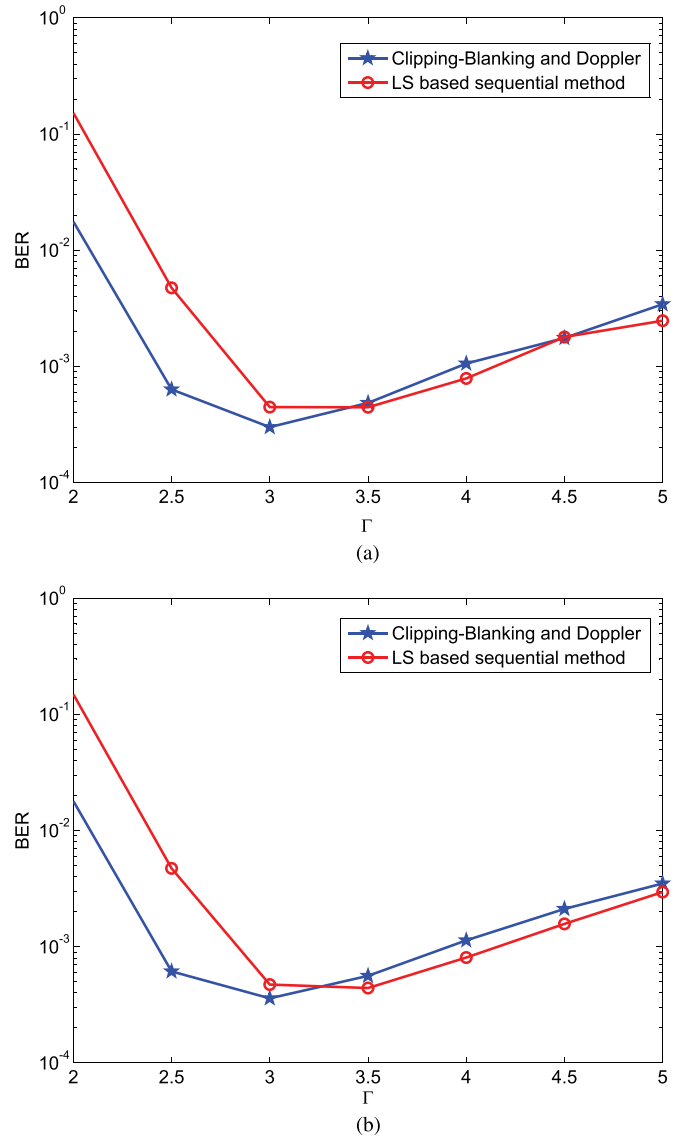


Fig. 4. BER performance as a function of threshold value. (a) $SIR = -10$ dB, $SNR = 15$ dB, $p = 0.02$. (b) $INR = 26$ dB, $SNR = 15$ dB, $p = 0.02$.

is evaluated where Doppler compensation is applied after impulse noise mitigation. In the second receiver, the impact of the threshold to mark the impulsive noise positions is evaluated for the LS-based sequential method. Based on the empirical experience with plots such as Fig. 4, the clipping threshold is set as $\Gamma_{clip} = 3$, and the blanking threshold is $1.5\Gamma_{clip}$ for the clipping-blanking method, whereas the position-marking threshold is set at $\Gamma = 3.5$ for all LS-based methods for the performance evaluation later on.

C. Performance Comparisons

We compare the following nine receiver configurations. The first group of receivers does not compensate the Doppler effect.

- 1) *No Compensation*: This receiver bypasses both Doppler shift compensation and impulsive noise mitigation.
- 2) *Doppler compensation alone*: This is the receiver from the work in [28] that performs Doppler shift compensation but no impulsive noise mitigation.

- 3) *Clipping-blanking alone*: This is the receiver from the work in [6] that mitigates the impulsive noise mitigation through the clipping-blanking operation but bypasses the Doppler shift compensation.

The second group of receivers is constructed from the existing literature.

- 4) *Clipping-blanking and Doppler*: This receiver uses the clipping-blanking method to suppress the impulsive noise as in [6], followed by the Doppler shift compensation as in [28].
- 5) *CS-based sequential method*: This receiver performs the Doppler shift compensation as in [28], followed by impulsive noise suppression based on CS [38].
- 6) *CS-based joint method*: This is the receiver from the work in [38] that performs joint Doppler shift and impulsive noise estimation based on CS.

The third group of receivers is the proposed receivers of this paper.

- 7) *LS-based sequential method*: This receiver performs the Doppler shift compensation as in [28] followed by impulsive noise suppression based on the LS formulation, as discussed in Section III-A.
- 8) *LS-based joint method*: This is the proposed method in Section III-A with joint Doppler shift and impulsive noise estimation.
- 9) *Iterative method*: This is the proposed iterative receiver in Section III-B based on an LS formulation. The stopping criterion is to set the maximum number of iterations to be one or two.

In addition, two performance benchmarks are included.

- 10) The impulsive noise positions are known to the LS-based joint method. This serves as a lower bound to the performance of the LS-based joint method where the impulsive noise positions have to be estimated.
- 11) Doppler shift compensation is carried out on the simulated data in the absence of impulsive noise. This is the performance lower bound for all the impulsive noise mitigation approaches.

Figs. 5–7 show the BER performance as a function of SNR, with SIR = −5 dB, −10 dB, and SIR = −20 dB, respectively. Fig. 8 shows the BER performance as a function of SNR but with INR = 26 dB. We have the following observations from Figs. 5–8.

- The receiver with no compensation and the receiver with only impulsive noise mitigation perform poorly. The receiver with only Doppler compensation works well when SIR = −5 dB in Fig. 5 but quickly deteriorates to an unacceptable level when SIR decreases to −20 dB in Fig. 7.
- The receivers based on the CS approach do not work as well as the LS-based approach. This could be due to the limited number of null subcarriers, since, on average, there are 20 impulsive noise samples per block. A further comparison of the CS and LS methods will be pursued in Section IV-D.

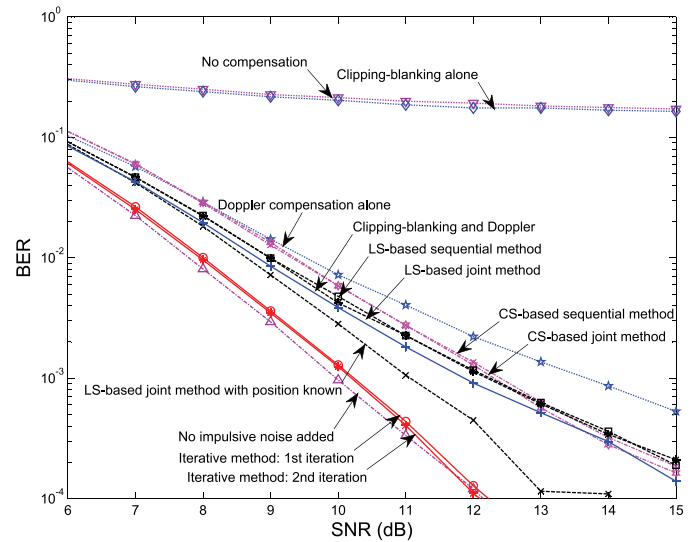


Fig. 5. BER performance of receivers with and without interference cancellation at SIR = −5 dB.

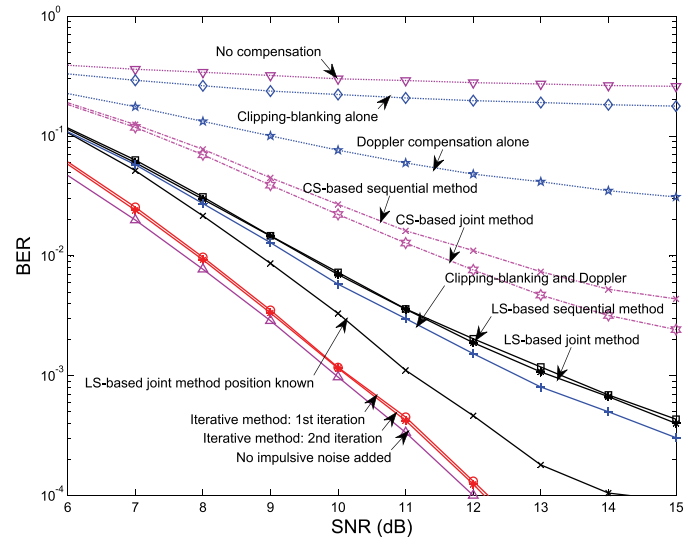


Fig. 6. BER performance of receivers with and without interference cancellation at SIR = −10 dB.

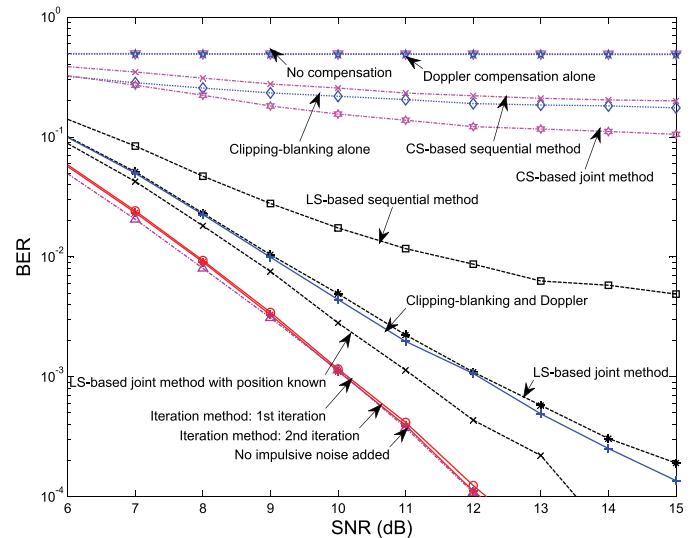


Fig. 7. BER performance of receivers with and without interference cancellation at SIR = −20 dB.

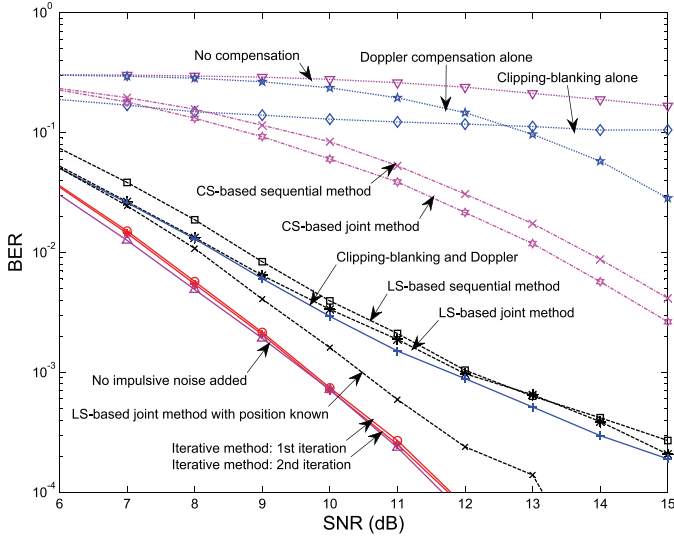


Fig. 8. BER performance of receivers with and without interference cancellation at INR = 26 dB.

- The LS-based sequential and joint methods have similar performance when SIR is -5 and -10 dB; however, the joint method is much better when SIR decreases to -20 dB in Fig. 7.
- The receiver with clipping–blanking impulsive noise mitigation followed by Doppler compensation is very effective. Its performance is close to that of the LS-based joint method in all settings of Figs. 5–8, whereas its complexity is much lower. Hence, this receiver is particularly appealing in the family of noniterative receivers.
- Compared with the benchmark where the impulsive noise positions are known, the LS-based joint method suffers from $1\sim 2$ dB performance loss at the BER level of 10^{-3} in Figs. 5–8.
- Most impressively, the iterative receiver improves the system performance significantly with only one iteration and outperforms all noniterative receivers. Further iterations have negligible gains. Note that the iterative receiver approaches the benchmark performance in the absence of impulsive noise (within 0.2 dB at the BER of 10^{-3} in Figs. 5–8). This suggests that the iterative receiver with one iteration can almost completely remove the effect of impulsive noise in our simulation settings.

D. Impact of the Number of Null Subcarriers

There are 96 null subcarriers in the signal design, whereby 48 of them are mixed with the data subcarriers, and 24 of them are on each side of the frequency band. Fig. 9 further compares the CS- and LS-based joint methods with different numbers of null subcarriers, where SIR = -10 dB, and $p = 0.02$. In the case of 60 null subcarriers, those 48 null subcarriers in the middle and 12 null subcarriers on each side are used. In the case with 144 null subcarriers, the extra 48 data subcarriers are converted to null subcarriers, and hence, the data rate is reduced to $R = (1/2) \cdot (((672 - 48) \cdot \log_2 4) / (T + T_g)) = 2.50$ kb/s. We can observe that the decoding performance of the CS-based

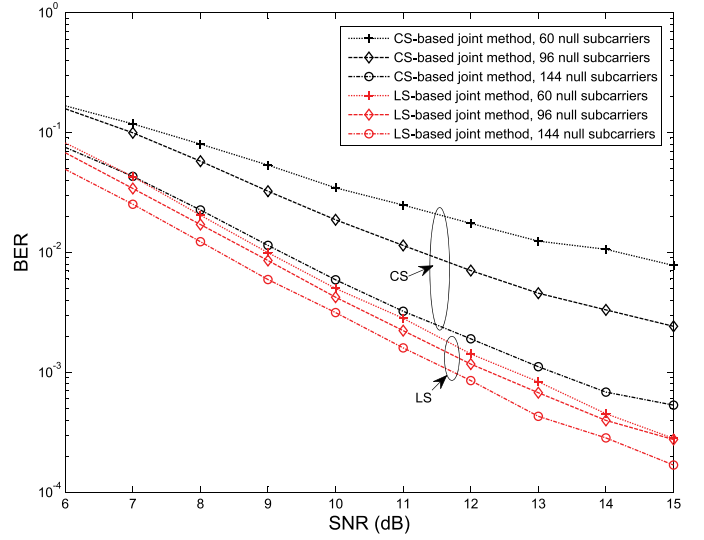


Fig. 9. BER performance with different numbers of null subcarriers, SIR = -10 dB, and $p = 0.02$.

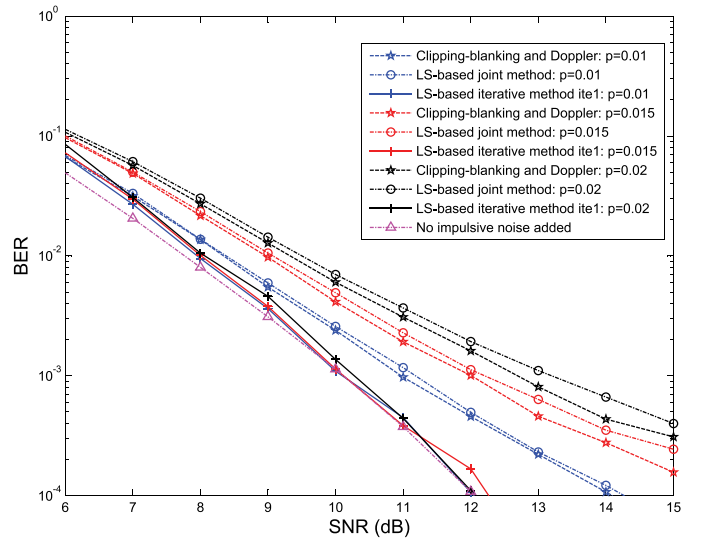


Fig. 10. BER performance with different impulsive noise probabilities, SIR = -10 dB.

joint method is considerably improved as the number of null subcarriers increases, whereas the improvement for the LS-based joint method is moderate. In all cases, the LS-based joint method achieves better performance than the CS-based joint method, and the gap is more pronounced when the number of null subcarriers decreases. This performance advantage is due to the utilization of the large-amplitude property of impulsive noise in the LS-based method.

E. Impact of the Probability of Occurrence of Impulsive Noise

Fig. 10 shows the performance of three receivers: the clipping–blanking and Doppler method, the LS-based joint method, and the iterative method (with one iteration), at different probabilities of occurrence of impulsive noise, where SIR = -10 dB. As p increases from 0.01 to 0.02 , the performance of

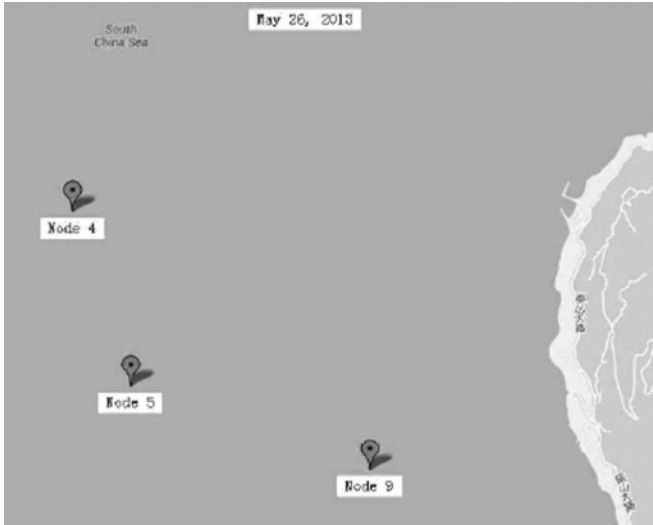


Fig. 11. Locations of the sources (node 4 and node 9) and the destination (node 5).

the noniterative methods deteriorates. However, in these tests, the iterative receiver can approach the performance benchmark when the impulsive noise is absent.

V. PERFORMANCE RESULTS WITH EXPERIMENTAL DATA

The Underwater Sensor Network Laboratory at the University of Connecticut participated in a joint experiment led by the National Sun Yat-sen University, at the sea near Kaohsiung city, Taiwan, on May 22–28, 2013. Tests with different purposes have been conducted; for example, performance results for a network-coded cooperation test have been reported in [49], and performance results for two adaptive modulation and coding tests have been reported in [50]. Inspecting the recorded waveforms, impulsive noise samples were observed in some of the received signals, and this fact has been mentioned in both [49] and [50]. There were significant shipping activities near the experimental site, and we suspect that the impulsive noise samples are due to the shipping activities.

A. Channel Conditions Corresponding to the Data Sets

In this paper, we look into the recorded data files corresponding to 17 data bursts, where three nodes were involved, as shown in Fig. 11. The distance between node 4 and node 5 was about 1550 m, and the distance between node 5 and node 9 was about 2188 m. Node 4 and node 9 worked as the transmitter, whereas node 5 worked as the receiver. The water depth was about 26 m, and the modems were about 6 m below the surface buoy. The maximum transmission power of the modems is around 20 W, and a -3 -dB power setting was used when collecting the data sets, leading to transmission power of around 10 W. Anchored to the sea bottom, all the nodes were stationary with small motions due to underwater currents and surface waves.

The channel delay profiles are shown in Fig. 12, where the channel delay spread is about 20–30 ms. Certainly, the channels in this experiment have a delay spread less than the guard inter-

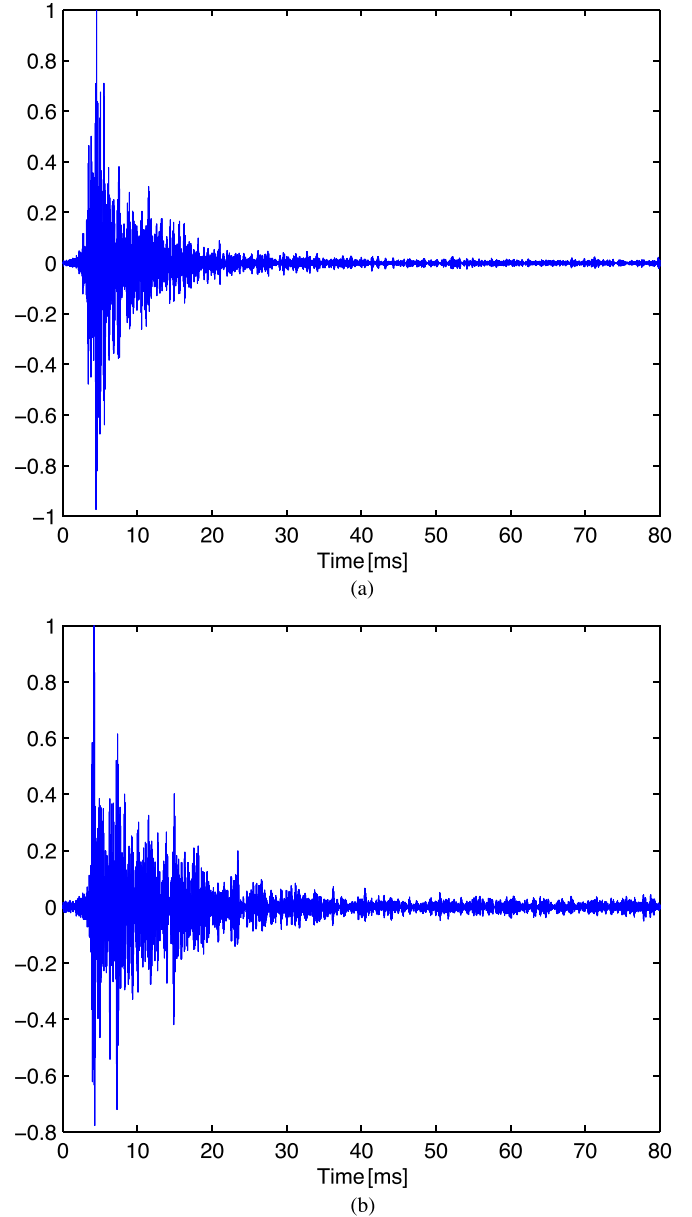


Fig. 12. Channel profile obtained by the hyperbolic-frequency-modulated preamble matching. (a) From node 4 to node 5. (b) From node 9 to node 5.

val. Fig. 13 shows the estimated channel scattering functions, where one can observe that the channels show nonnegligible Doppler drifts. Since the nodes were stationary and the Doppler shift is found to be varying within half the subcarrier spacing, the resampling operation was not used to process these data sets, i.e., $\hat{a} = 0$. Doppler shift compensation is done jointly with impulsive noise estimation on a block-by-block basis, and the performance results of different methods will be reported next.

B. Performance After Block Combining

For each recorded file, there are 80 OFDM blocks with the parameters specified in Table I. For a total of 17 recorded data files, there are 1360 OFDM blocks. Not all the blocks are contaminated by impulsive noise samples. After passband-to-baseband downshifting and LPF, the time-domain waveform of

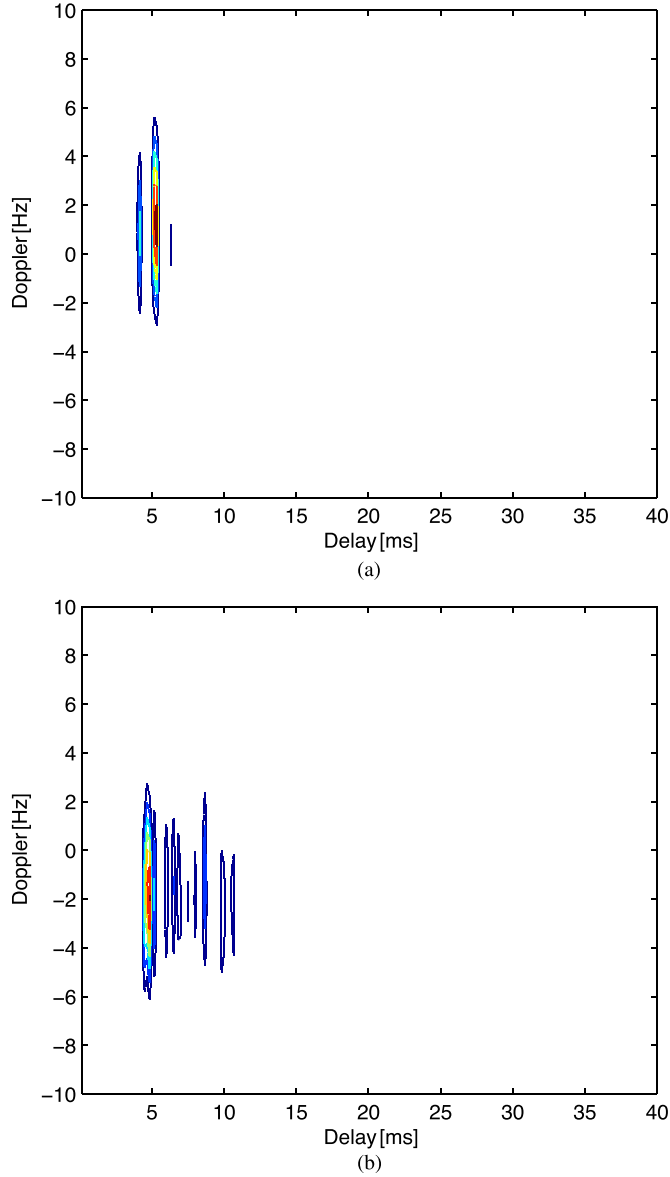


Fig. 13. Estimated scatter function in the experimental test. (a) From node 4 to node 5. (b) From node 9 to node 5.

each block is inspected. A block is viewed to be contaminated by impulsive noise samples if there are large-amplitude samples identified. This way, we kept 26 blocks with large-amplitude impulsive noise samples, which is about 2% of the total blocks. The baseband waveforms for three blocks in the presence of impulsive noise are shown in Fig. 14, where the signals were affected with different degrees of severity of the impulsive noise.

For each block, the transmitted information symbols are the same. Here, we combine multiple blocks for decoding and plot the BER performance as a function of the number of combined blocks. In case of combining N blocks, there are $C_{26}^N = 26!/(N!(26-N)!)$ different combinations over which the BERs will be averaged. The blocks are combined as follows. Consider the ν th block to be combined. After the Doppler shift compensation and impulsive noise mitigation as in (19), denote the output on the m th subcarrier as $\hat{z}_\nu[m]$. Moreover, denote the channel estimate on the m th subcarrier as $\hat{H}_\nu[m]$.

For the decoding purpose, one can use an equivalent channel input–output relationship, i.e.,

$$\hat{z}_\nu[m] = \hat{H}_\nu[m]s[m] + \eta_\nu[m] \quad (25)$$

where $\eta_\nu[m]$ contains not only ambient noise, residual intercarrier interference, and residual impulsive noise samples but also the noise due to the channel estimation error. A maximum ratio combining (MRC) approach can be used to combine N blocks together as [51, ch. 9]

$$\hat{z}_{\text{mrc}}[m] = \frac{\sum_{\nu=1}^N \hat{H}_\nu^*[k] \hat{z}_\nu[m]}{\sqrt{\sum_{\nu=1}^N \hat{H}_\nu^*[m] \hat{H}_\nu[m]}}. \quad (26)$$

Define $\hat{H}_{\text{mrc}}[k]$ as the equivalent channel gain, i.e.,

$$\hat{H}_{\text{mrc}}[m] = \sqrt{\sum_{\nu=1}^N \hat{H}_\nu^*[m] \hat{H}_\nu[m]}. \quad (27)$$

Plugging (25) into (26), the channel input–output relationship for the MRC output is

$$\hat{z}_{\text{mrc}}[k] = \hat{H}_{\text{mrc}}[k]s[k] + \eta_{\text{mrc}}[k] \quad (28)$$

where $\eta_{\text{mrc}}[m]$ is the combined noise. The SNR at the MRC output is the sum of the SNRs of the individual subchannels, leading to improved decoding performance [51]. The rationale behind this approach to compare different receivers is as follows: 1) The SNR, SIR, and INR are not well defined and hard to quantify for the collected data blocks; 2) the number of data blocks is limited; and 3) the combination of multiple blocks resembles that of a multihydrophone receiver, where the signals from multiple hydrophones are combined. Plotting the system performance as a function of the number of hydrophones at a receiver is a common practice for UWA communications [33].

The decoding performance of different receivers is shown in Fig. 15. We also have recorded noise samples when the signal was not transmitted. To generate more blocks for performance evaluation, we add the impulsive noise samples from the same experiment to the data blocks without impulsive noise. The decoding performance of different receivers is shown in Fig. 16, where the benchmark performance when the impulsive noise samples are not added is also included. The SNRs and SIRs are different for different blocks. A rough estimation suggests that the SIRs are fluctuating around -8 dB, whereas the SNRs are fluctuating around 10 dB for most of the blocks.

We have the following observations from Figs. 15 and 16.

- The receiver with no compensation performs poorly. The receiver with only impulsive noise mitigation marginally improves, whereas the receiver with only Doppler compensation considerably improves. Hence, for these data sets, the impact of Doppler shift is much more severe than that of the impulsive noise, confirming that Doppler compensation cannot be ignored even when the communication nodes were stationary.
- When both the Doppler shift compensation and impulsive noise mitigation are performed, either sequentially or jointly, the receiver performance significantly improves.

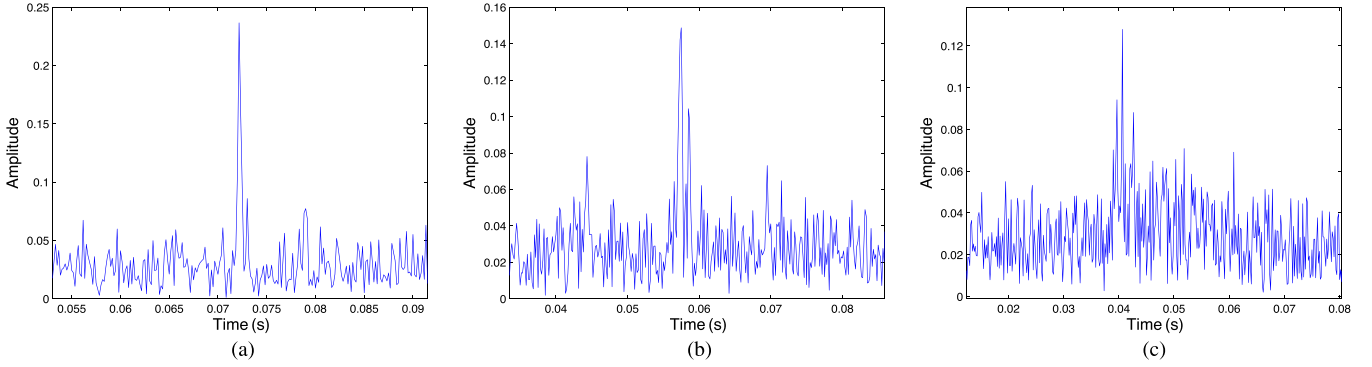


Fig. 14. Baseband waveforms of OFDM blocks with impulsive noise.

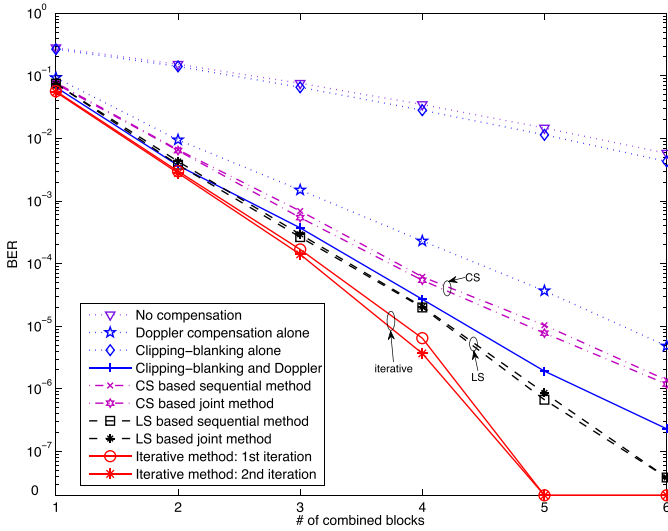


Fig. 15. BER performance based on the data blocks with impulsive noise samples.

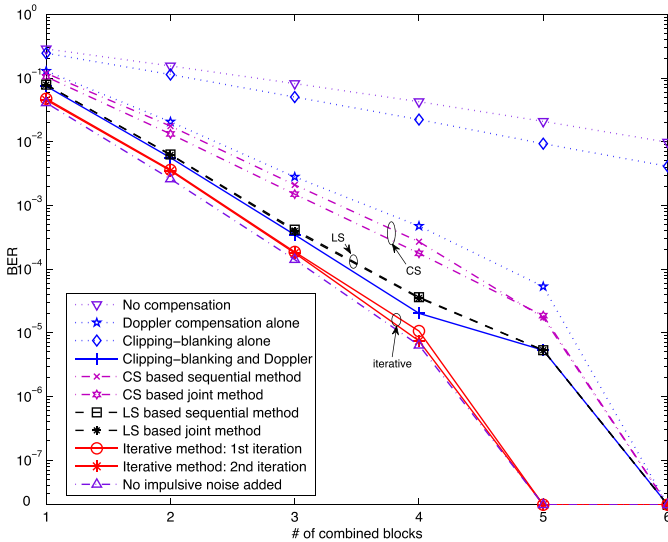


Fig. 16. BER performance based on the data blocks with recorded impulsive noise samples added.

Again, the receiver with clipping-blanking impulsive noise mitigation followed by Doppler compensation is very effective. The proposed receivers based on the LS formulation slightly outperform the clipping-blanking method together with Doppler compensation in Fig. 15 but have

similar performance in Fig. 16. The sequential Doppler compensation and LS-based interference mitigation have almost identical performance with the joint search.

- The receivers based on the CS approach do not work as well as the LS-based approach, where 96 null subcarriers are used.
- With one iteration, the system performance from the iterative receiver is obviously improved. Further iterations do not help any more. The iterative receiver approaches the benchmark performance; hence, the effect of impulsive noise can be almost completely removed for the testings in Fig. 16.

Overall, the observations from the experimental data are consistent with the simulation results, although the BER plots are generated against different variables.

VI. CONCLUSION

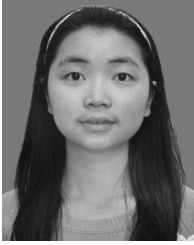
In this paper, we have studied impulsive noise mitigation for UWA OFDM systems. We present an LS-based approach, which first detects the positions of the impulsive noise in the time domain and then estimates the impulsive noise samples jointly with the Doppler shift based on the measurements on OFDM null subcarriers. Based on the available channel estimate and tentative data symbol decisions, an iterative receiver was further developed. The simulation noise was generated based on a GM model, and the model parameters were derived from the real data. Both simulation and experimental results demonstrated that the proposed receivers are very effective. In particular, within the family of noniterative receivers, one existing receiver with clipping-blanking impulsive noise mitigation followed by Doppler compensation is appealing due to its effectiveness and low complexity, whereas the proposed iterative receiver with one iteration can approach the benchmark performance obtained in the absence of impulsive noise, suggesting that impulsive noise samples can be satisfactorily removed in practical UWA OFDM systems.

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